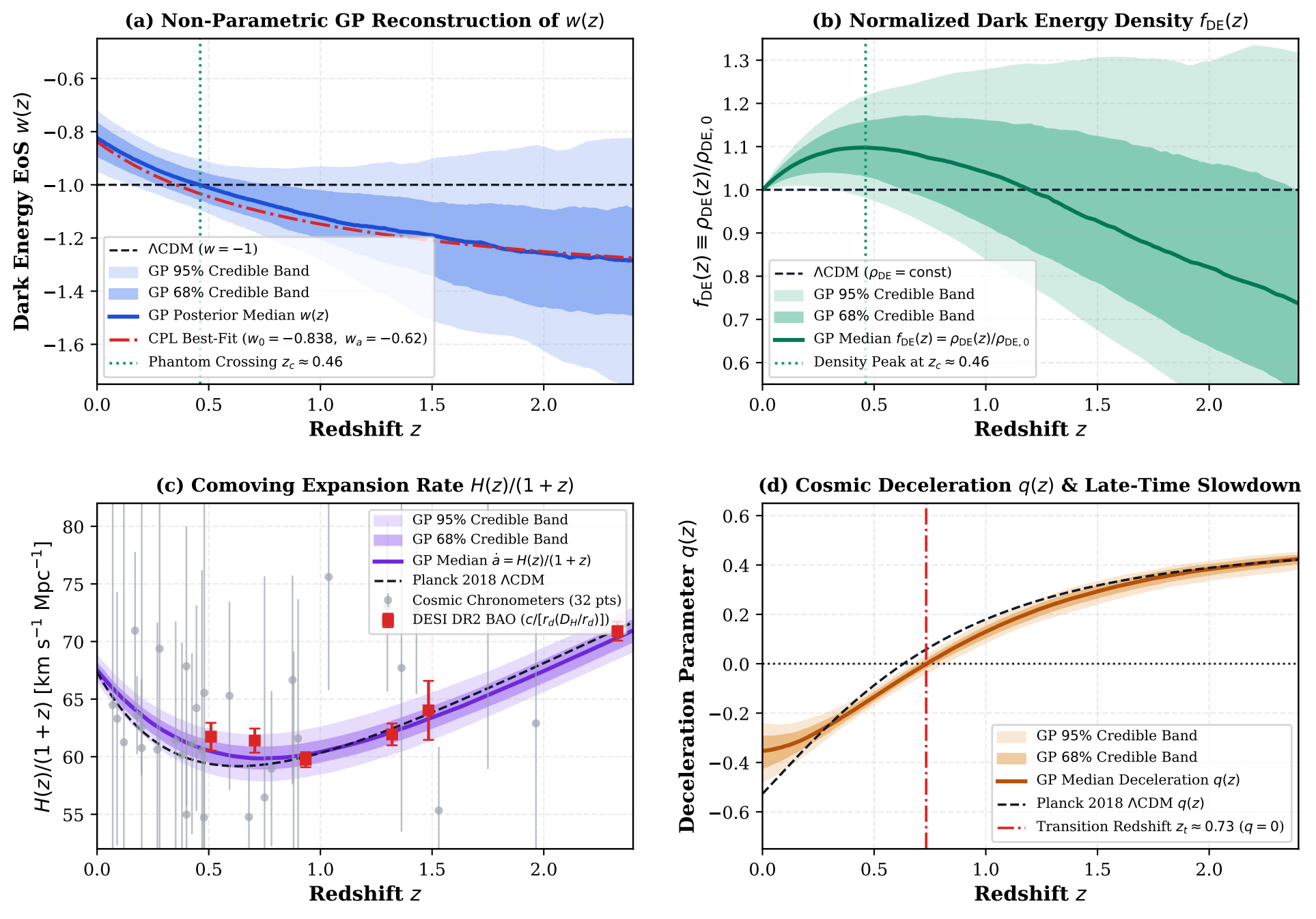

(a) Non-Parametric GP Reconstruction of $w(z)$
Dark Energy EoS $w(z)$
Redshift z
ΛCDM ($w = -1$)
GP 95% Credible Band
GP 68% Credible Band
GP Posterior Median $w(z)$
CPL Best-Fit ($w_0 = -0.838,\ w_a = -0.62$)
Phantom Crossing $z_c \approx 0.46$
(b) Normalized Dark Energy Density $f_{\rm DE}(z)$
$f_{\rm DE}(z) \equiv \rho_{\rm DE}(z)/\rho_{\rm DE,0}$
Redshift z
ΛCDM ($\rho_{\rm DE} = \text{const}$)
GP 95% Credible Band
GP 68% Credible Band
GP Median $f_{\rm DE}(z) = \rho_{\rm DE}(z)/\rho_{\rm DE,0}$
Density Peak at $z_c \approx 0.46$
(c) Comoving Expansion Rate $H(z)/(1+z)$
$H(z)/(1+z)$ [km s^{-1} Mpc^{-1}]
Redshift z
GP 95% Credible Band
GP 68% Credible Band
GP Median $\dot{a} = H(z)/(1+z)$
Planck 2018 ΛCDM
Cosmic Chronometers (32 pts)
DESI DR2 BAO ($c/[r_d(D_H/r_d)]$)
(d) Cosmic Deceleration $q(z)$ & Late-Time Slowdown
Deceleration Parameter $q(z)$
Redshift z
GP 95% Credible Band
GP 68% Credible Band
GP Median Deceleration $q(z)$
Planck 2018 ΛCDM $q(z)$
Transition Redshift $z_t \approx 0.73$ ($q = 0$)